Impact of Pharmacist Clerking on Right-First-Time Admission Prescribing in an Acute Medicine Unit.

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Abstract

This paper is expected to be the first of several looking at different aspects of medicines management within the Acute Medical Unit stream of patients admitted to University Hospital Galway. This first paper attempts to critique and remodel the system for assuring a high quality of admission medication histories. In Plan-Do-Study-Act (PDSA) vernacular, it represents round one of the improvement cycle. The most urgent, important and eminently fixable problem has been identified to achieve quick wins and a massive improvement in yield to the overall process.

The overall aim of this series of medicines management projects is to optimise system design in the pursuit of excellent medicines information delivery throughout the admission, inpatient, transfer and discharge process to achieve **seamless pharmaceutical care** across multiple interfaces. We want patients journeys to begin on day 1 with a **complete and accurate** medication history to better inform differential diagnosis, prognosis and treatment plan.

At the end of this first cycle of improvement we have achieved a reduction in median medication admission anomalies from 3 to 0. As far as patients are concerned, this translates to a reduction in the number of patients affected by medication admission anomalies from 92% to 2%. Stakeholder feedback has been very positive from physicians and nurses. The expectation based on international best practice is that this will have a major benefit on harm reduction for patients also.

In fact, the new system design has been so successful that it has required us to rethink how we measure what is now a rare event. With this success we have unearthed our next niggle. That niggle is unearthing demand, load, capacity activity data that will inform us how best to match our resources with incoming demand. (286 words)

Keywords

Healthcare; Acute Medicine; Medication; Errors; Harm; Medicines Reconciliation; Pharmacist; Clerking; Admission; Safety;

Context

The World Health Organisation (WHO) estimates the cost of medication harm at $42,000,000,000 US annually (1). Whilst the financial costs are very significant, the human costs of omitting vital treatment, under-treating a patient or exposing them to unnecessary toxicity are also very significant.

One particular area of focus by governments and the WHO is medicines reconciliation (1). This is the process of “making sense” and “finding the true medication record” for a patient when they cross a care interface e.g. from the general practitioner to the local emergency department. Not only was this a critical patient safety improvement but there was evidence that it made good business sense (2).

University Hospital Galway has formalised pharmacist-provided medicines reconciliation on admission for several years now. Since 2014, all local undergraduate medical students have been trained and examined on prescribing behaviours and this training includes structured formal medicines reconciliation. Unfortunately, a large volume of anecdotal evidence suggested that this training had not translated into different prescribing behaviours for the majority of prescribing interns.

In 2014, having dabbled in Lean, PDSA and Six Sigma, the author became aware of the Safety-Flow-Quality-Productivity (SFQP®) model of Health Care Systems Engineering (HCSE) developed by Simon Dodds. This robust, scientific and measured approach to quality improvement in complex adaptive healthcare systems provided the basis for a scientific approach to process improvement where medicines reconciliation was concerned.

Purpose

The purpose of this first phase of improvement was to address a recurring “niggle”, one that pre-occupied all hospital clinical pharmacists and medicines-focused consultant physicians: That of disruptive, confusing and inaccurate medication histories at the point of admission in the Acute Medical Unit in University Hospital Galway.

Method

The Safety-Flow-Quality-Productivity framework was used as the basis for prioritising deliverables in small improvement chunks in a time-phased manner. In the SFQP® framework, safety always comes first, and this was particularly relevant given that our “right-first-time" yield was known to be extremely poor. Anecdotally there was a large body of evidence that suggested that improving safety by getting medicines right-first-time would likely lead to better flow, quality and productivity.

More information about the SFQP® framework can be found at: https://www.improvementscience.co.uk/blog/?p=3919.

Stage 1: Map

Figure 1: System map describing patient and information flows

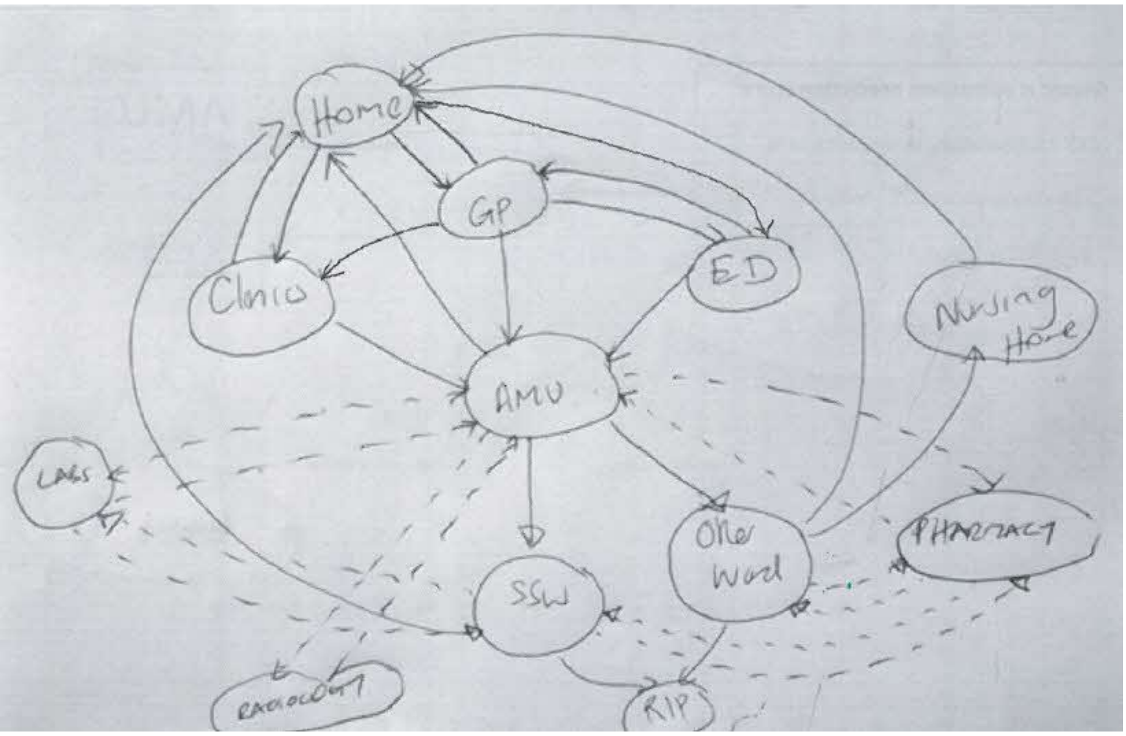
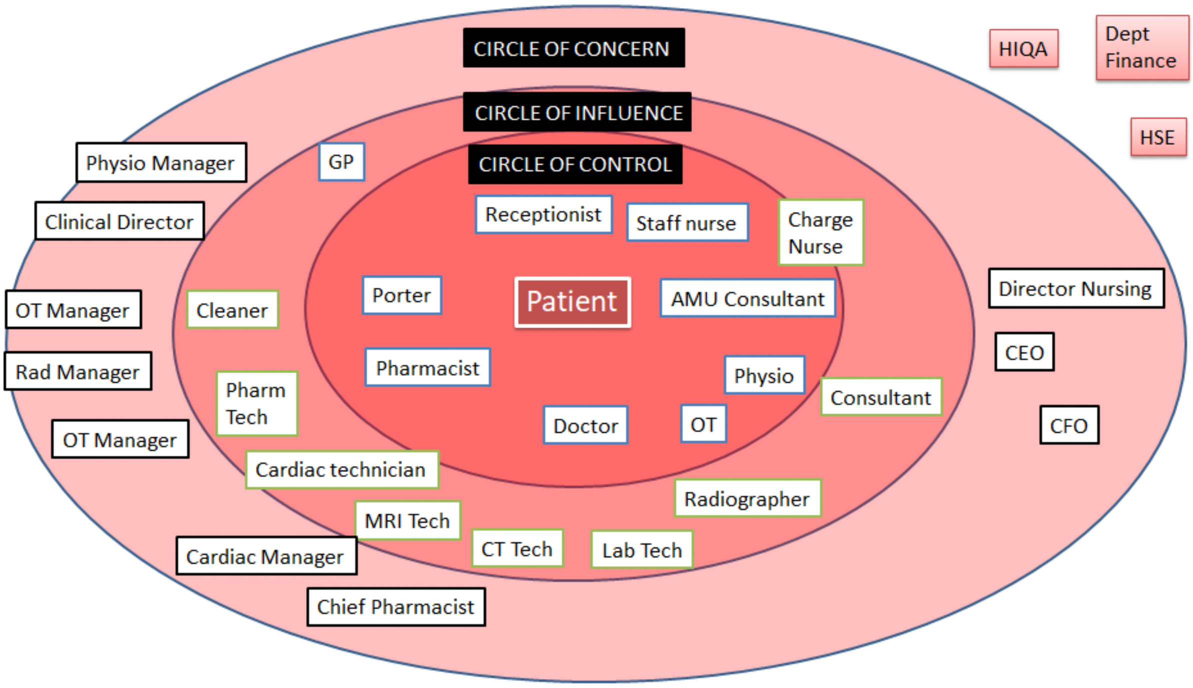


Figure 2: Getting to the heart of the matter with 4Ns: Nuggets, Niggles, Nice Ifs and No-Nos of the group

Why is this important? Why should you care? (feedback from customers)			
Nuggets (Past & Present)		**In the future it would feel NICE IF…**	
Appreciated	AMU team looking to use pharmacists' skills to streamline a safer admission and discharge process	Confident	We could be confident that a **complete and accurate medication history was fully documented** on the drug chart **and right-first-time**
Optimistic	System redesign will almost eliminate medication admission anomalies in phase 1 of project	Appreciated	If hospital management recognized the value in getting medicines right first time for patients and staff
Confident	Improvement science training will act as an enabler given AMU/pharmacy teams competencies in this regard	Fully utilised	If pharmacists were also involved with assessment and clerking of LMWH for DVT prophylaxis (consultants request)
Niggles (Past & Present)		**No-Nos that will not happen in future**	
Frustrated	Practical teaching to transition year students has had a very limited impact on medication admission anomies	Angry & upset	If a patient suffers permanent harm as a result of medication admission anomalies
Concerned	Omission errors at admission are of the "serious harm" category and in the 15-25 HSE risk matrix zone	Disappointed & dejected	If a clerking pharmacist were to introduce new errors following system change

Phase 1 of our Medicines Reconciliation (MedRec) redesign project is the admissions interface. Too many patients are at risk of medication harm as a result of incomplete/inaccurate drug charts which are written up by insufficiently trained and new to the hospital non-consultant hospital doctors.

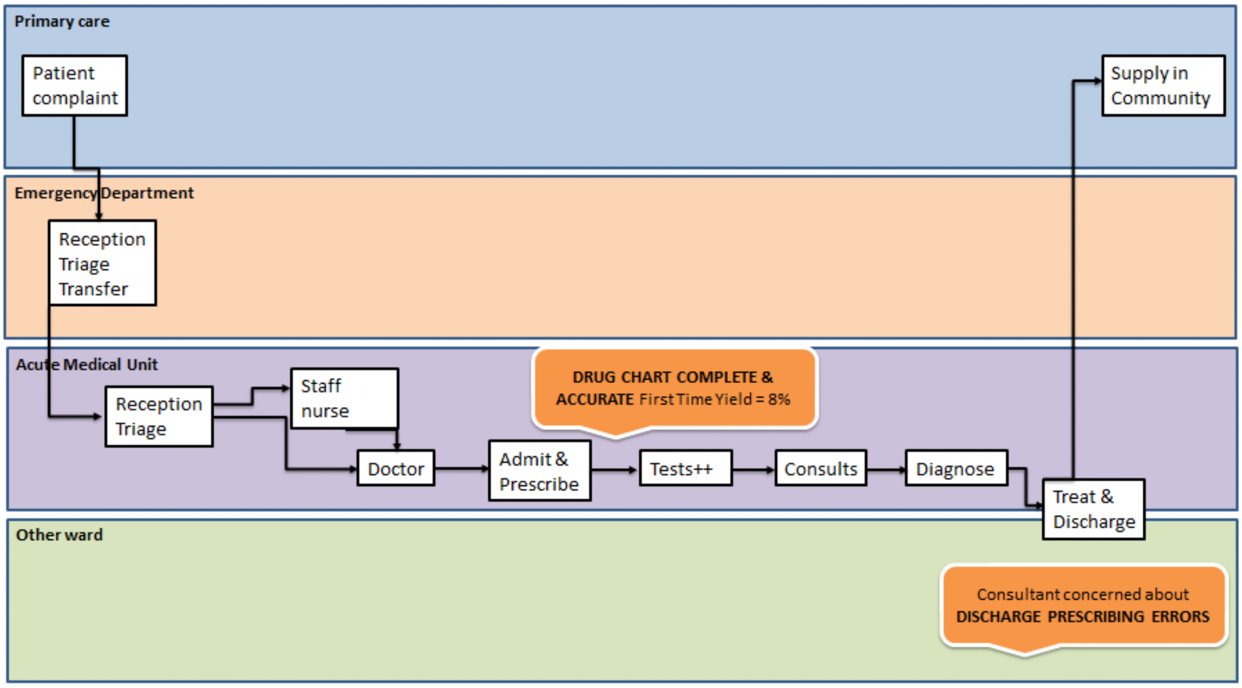
Figure 3: Stakeholder map.



The stakeholder map acted as a gentle reminder that phase 1 of the improvement needed to be geared toward **"looking within" our team** to determine how we could directly influence the accuracy of medication admission histories by acting within our circle of control.

Solutions were identified beyond our circle of control such as electronic prescribing that linked with the primary care interface. However, the niggle-o-gram® indicated that this not where the low hanging fruit lay and that **to wait for a technological solution would do patients a grave injustice** while errors continued to happen on our watch.

Figure 4: The Process

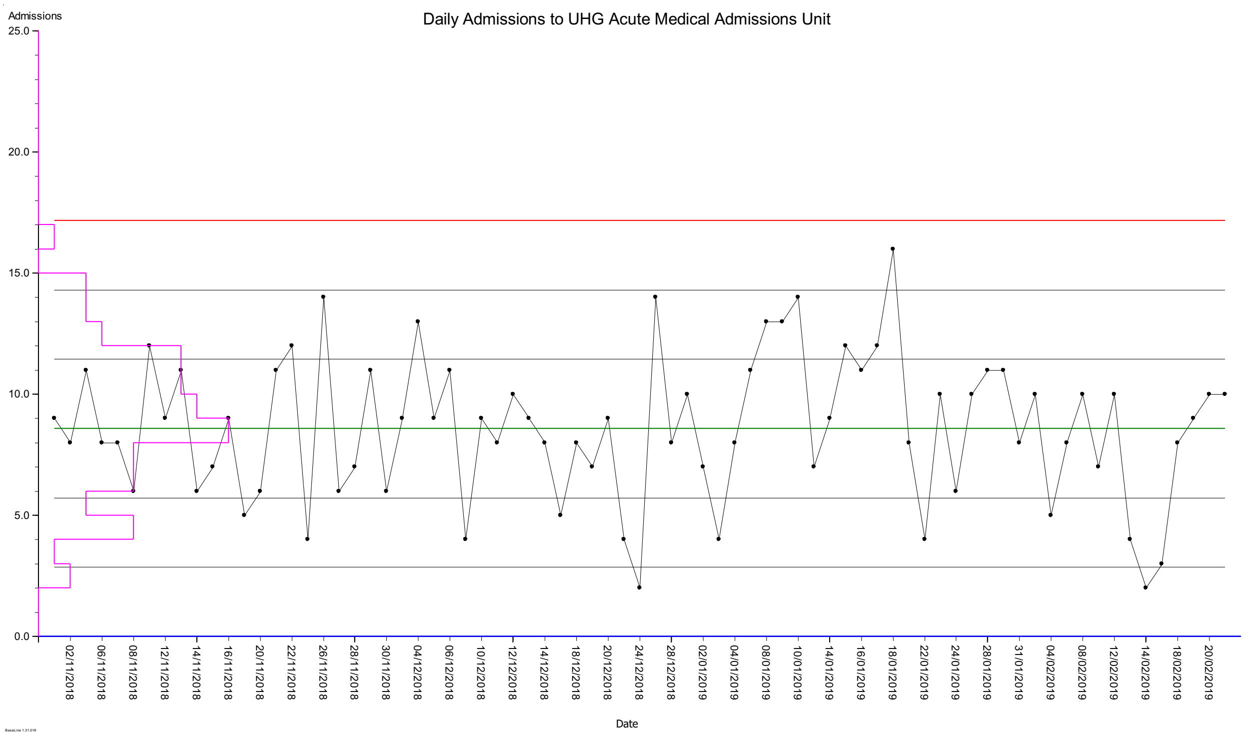


Whilst the conversation that started this project was the consultants concern with discharge prescribing, it was decided that we needed to **investigate failure further up-stream and fix the admissions process as the first priority.** The traditional model of working relied upon quality inspection and correction, rather than "built-in quality". In other words, pharmacists intervened after prescribing errors or omissions were made and tried to put out the fires. Despite extensive teaching through the medical school the errors persisted and no real impact had been observed on yield for complete and accurate prescribing first-time.

Stage 2: Measure

Over the period in question a range of 25 to 30 patients attend daily for review. On average 8.6 patients per day are admitted (Fig 5. Sigma=2.9, Median=9) and the process is stable. In this same period, it was found that 26% of patients were admitted and 74% were discharged.

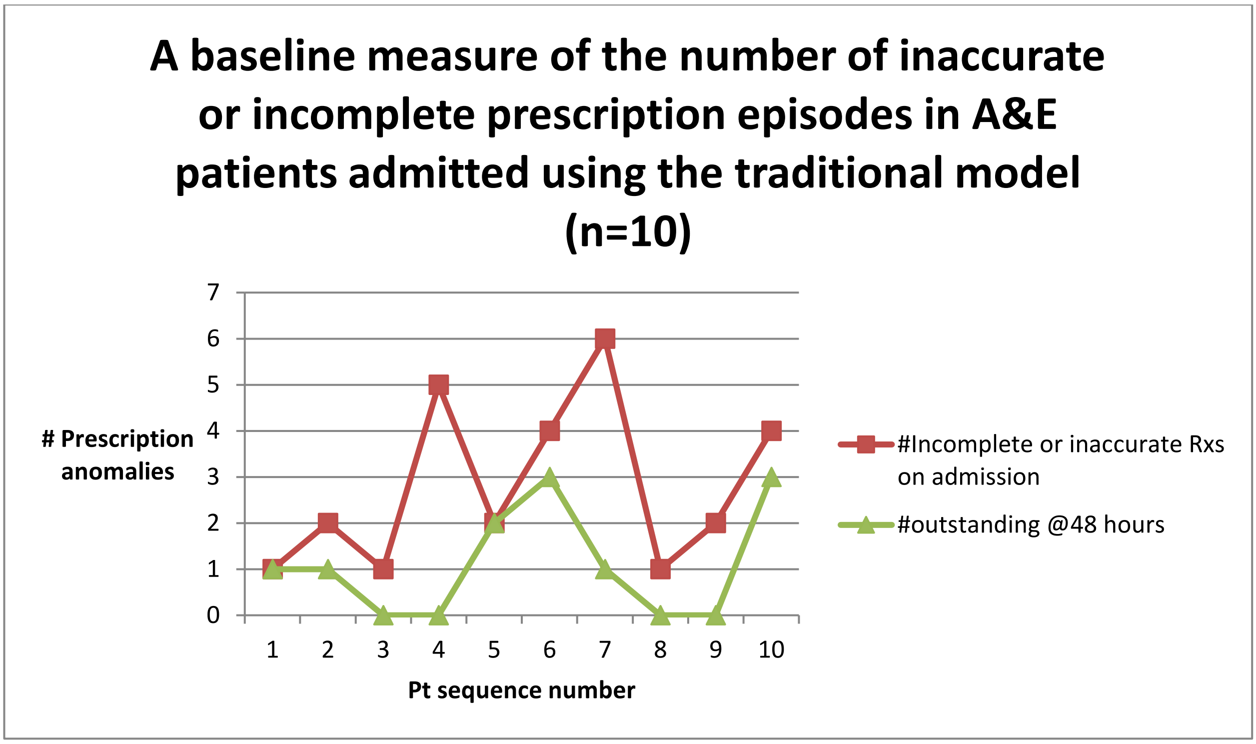
Figure 5: Daily admissions to the Acute Medical Unit at UHG



This information is useful in that it informs the design team what level of patient demand is predicted on any given day. Unfortunately, not all admissions are the same. so the work (load) required to accurately clerk every patient's medicines will be dependent on several factors, including the number and nature of the medicines a patient is taking.

With this data at hand, a baseline measure was conducted in a small number of patients in order to give some appreciation for the scale and frequency of the admission medication inaccuracy problem using a traditional NCHD "admit-prescribe" model. Patients were selected for audit if they were admitted and fulfilled in-house criteria for being either at intermediate or high risk of medication harm according to Galway University Hospitals guidelines.

Figure 6: Baseline measure of the number of incomplete or inaccurate prescription episodes per patient admitted using the traditional NCHD admit-prescribe model of medicines reconciliation.



Two major niggles were identified here. The first was that there was at least one prescription anomaly in every patient audited. The second niggle was that despite clear written communication of prescription anomalies 11 out of 28 were still outstanding at 24 hours (with no explanation).

The traditional medication admission model involves a medication history being taken by a non-consultant hospital doctor (NCHD) without a documented standard process. It typically involves a GP letter and a patient interview where this is feasible. The 9-point checklist (below) is rarely annotated and when this is the case it is commonly found that the drug chart does not contain a full and accurate history of baseline medicines use as per Fig 6 above.

The clinical pharmacy lead decided that at this point there was no need to investigate further or “measure more”. In-house data using the PharmaBOT app confirmed that admission medication history anomalies was the single largest medication safety issue documented (3). When combined with the evidence from *Medication Without Harm: WHO's Third Global Patient Safety Challenge* (1) we knew we needed to act now and perfect later.

Stage 3: Model

3.1 Diagnosis

Figure 7: A "5-whys analysis" was completed in order to better diagnose the root-cause of the problem and hopefully identify a design that would minimise recurrence of the problem.

5-Why Analysis

Problem: Too many incomplete or inaccurate drug charts prepared during the admission process

Why 1	Why 2	Why 3	Why 4	Why 5	Root Cause	Recurrence Prevention
Patient drug chart only includes SOME of the patient's baseline medications	Admission history is inaccurate	Admitting prescriber did not follow standard MedRec process	Did not believe it was important	50% of interns are NOT trained in this process (not NUIG students)	Lack of awareness & training together with large training pool	Senior clinical pharmacist as most knowledgeable medicines reconciliation operator to write drugs up on admission. Will also reduce inter-operator variation in process.
				Other things more important	Lack of awareness and time poor	
		Prescibers product knowledge insufficient	Limited experience prescribing drugs and inaedquate knowledge of drug/product/strength	Product familiarisation is not part of their training program	Poor skill mix secondary to inadequately trained operator	
	Prescriber "held" the drugs when the patient was admitted	Suspected drug toxicity	Side effect profile of drug consistent with reaction	Similar side effects reported & documented in clinical trials	Patient unable to tolerate drug or drug dose	Write all drugs including held drugs on drug chart and document intolerance
			Patient characteristics suggest likely to be sensitive to the drug	Reduced elimination secondary to renal, hepatic or drug interaction factors	Increased sensitivity secondary to poor drug combination choice or excessive dosing	Visually flag low therapeutic index drugs together with elimination and interaction issues on drug chart on day 1
			Previous drug toxicity in this patient	ADR Hx not reconciled completely before prescribing today	High variation in quality of documentation of adverse reactions	Better ADR documentation system at ADMIT stage

3.2 Design

Based on the 5-whys analysis below the following actions were identified for *"MedRec Redesign Mark 1"*.

1. Move the clinical pharmacist resource to the front of the admissions prescribing process.
2. Simultaneously order the drugs while the drug chart is prepared.
3. Expedite ordering from pharmacy.

During this time the mapping process proved invaluable. We knew that actions 1 to 3 outlined above would:

1. **Most importantly be a win for patients by optimising treatment outcomes through getting medicines right first time.**
2. Be **a win for doctors** because they could add more value with an accurate medication history already performed. It has been shown that incorrect drug histories can lead to cessation of required therapy, and failure to detect a drug-related problem (4).
3. Be **a win for doctors** because they would have **more time for high-value work.** Time saved by removing them from a process a more specialised and experienced practitioner could perform with a much greater level of accuracy and completeness.
4. Be **a win for nurses** because **medicines would be ordered for them** by the pharmacy team.
5. Be **a win for nurses** because **medicines would be ordered right first time** (less queries due to failure demand).
6. Be **a win for nurses** because their drug cupboards would not be **overflowing** with unnecessary drugs.

7. Be a win for **pharmacy because they would not need to supply what was already available.**
8. Additionally, be **a win for patients** because they would not have to **repeat their medication history** three times to three different practitioners who would document it with varying degrees of accuracy and understanding.

Through extensive stakeholder consultation using the 4N Chart® emotion-in-time map we could "get on the same page" when considering the change that needed to be made for the **betterment of patients' lives and the workers in the system.**

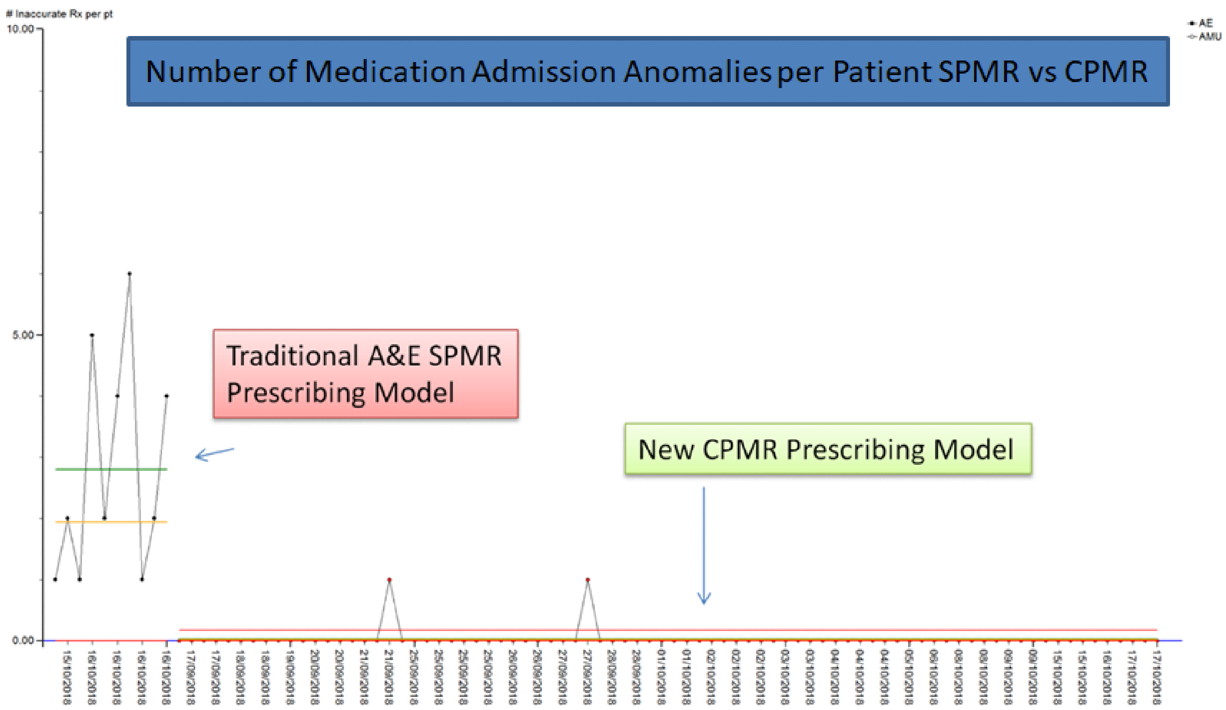
Stage 4: Modify

The problem was deemed both urgent and important to tackle. Following consultation with the Lead Antimicrobial Consultant and Charge Nurse based on the AMU, we commenced the pharmacist clerking workflow on 17th September 2018. The Consultant/Charge Nurse sponsor was key for the success of this project as they were able to utilise their circle of influence to get all other consultants, NCHDs and nurses on board with the process redesign.

The aim of this first phase of implementation was simply to clerk as many patients admitted through the Acute Medical Unit as possible. Capacity to admit 100% of patients using this new method was not considered critical. Redesign essentially required re-ordering of already established roles.

Stage 5: Monitor

Figure 8: Individuals chart grouped by admission system. The traditional A&E model of admission demonstrated a mean of 2.8 Medication Admission Anomalies (MAAs) per admission with large variation (sigma = 2.1). The new Pharmacist Clerking system clearly had a much lower MAA rate (mean = 0.025, sigma=0.045).



During this phase the clinical pharmacist was independently audited by another senior clinical pharmacist with 15 years' experience. There was found to be a 100% correlation.

Stage 6: Maintain

The new design is now firmly embedded in the culture of our AMU. However, as Taiichi Ohno father of the Toyota Production System (TPS) once said *"Having no problems is the biggest problem of all."(5).* While the pharmacy department and the AMU continue to deploy pharmacist clerking and monitor its performance we are now faced with a number of new challenges.

1. **How to measure the new system** now that the frequency of anomalies with pharmacist clerking is so low.
2. How to improve the pharmacist clerking: NCHD admission prescribing ratio (**more visibility of the new process is required for doctors on short rotations** and it needs to be driven by nurses who first see the patient).
3. **Training other pharmacists** so as to improve the capability of the hospital to reduce more medication admission anomalies and support the AMU during annual leave for example.
4. **Optimising capacity to perform clerking** given that it may not be entirely time neutral for the pharmacist.
5. **Embedding the new process into a redesigned drug chart** that clearly identifies clerked patients and supports the leanest possible process for doing and measuring demand, load, capacity and activity.

Addressing these issues, shall be the subject of the next phase of improvement in the AMU.

Reflections

This simple project has reinforced some very important fundamentals of change management. As they say in life, timing is everything. An enormous amount of time had been invested personally and within the department sharpening our improvement science skills. And whilst this had been underway in the background three years of groundwork had taken place building relationships beyond the four walls of pharmacy.

Like most things in life, serendipity has a role to play, and so does the importance of "making your own luck". In 2019, there was a new lead appointed for audit and quality in Galway University Hospitals. This same consultant had recently embarked on an RCPI sponsored quality improvement course with other key personnel in the hospital. The Galway University Hospitals audit manager had already successfully laid the groundwork for the teaching of quality improvement within the hospital and teams were beginning to find their wings. While the author had been waiting for the opportunity for spreading improvement science outside the four walls of pharmacy, finally it had arisen.

Suddenly, there was a newly established unit ripe for improvement with all the clinical leadership and will firmly in place. In essence, resistance to change was not going to be a problem.

The power of visual management reporting through PharmaBOT app deployment emphasised the importance of tackling the medicines admission problem as a priority. Together with the "right-first-time" principle it was realised that process disruption was required. The opportunity for massive improvement with one simple "left-shift" of "person in process" was too powerful to ignore. Critically there was no resistance amongst medics or nurses on the unit given the win-win-win scenario this change promised.

System mapping and a baseline audit ensured that we really were on the same page as to the extent and nature of the problem. The Nice-Ifs allowed us to clearly "frame a bullseye" around our goals.

Maintaining this new system will present with some challenges given that NCHDs rotate through the unit on training placements of limited duration. While local support amongst clerking pharmacists has been excellent, these pharmacists will require extra support for the extra responsibility they undertake in this role if we are to encourage other pharmacists to take on the role of clerking pharmacist. Paper 2 will explore these challenges in the context of ever-increasing capacity in an expanding AMU.

Lessons

1. Take the time to grow a willing coalition throughout the stages of a patient stream.
2. Identify an area where will is high and negativity low.
3. Measure just enough to provide that evidence base for change.
4. Involve important stakeholders in "strategic conversation" agreeing the NICE-IFS to frame the win-win-win solution.
5. Be very deliberate in constructing a language framework around what is best for patients and the workers in the system to minimise resistance.
6. Don't wait for everyone to jump on board. Form that multidisciplinary team of the willing coalition and get on with it.
7. Measure and mass-communicate

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Sinead Lydon has performed trojan work as our clinical pharmacist on AMU and worked very productively with Dr Louise Rabbitt. I sincerely hope this is the start of a beautiful relationship between NUIG School of Medicine and the Pharmacy Department in Galway University Hospitals. Thank you also to Tim Cronin our intern (Sinead's able deputy). Both Sean Collins and John Given have been wonderfully supportive as has Sheila Moran in her capacity as process validator.

Finally, I cannot thank enough all my "Quality in Action" colleagues who give their own time throughout the year to lead Quality Improvement Training in our hospitals. I am especially grateful to Tara Hannon for facilitating the running of the first two Flow improvement workshops. It is this community that sustains me in those more challenging moments when the organisational bureaucracy weighs me down!

A shout out also to Boston Scientific Galway who have been very open to share their Operational Excellence insights with our Improvement Science Community. We look forward to more learning later in the year.

2019 should be an exciting year where Improvement Science is concerned in Galway University Hospitals. I look forward to it immensely and the benefits it will bring to patient's and those absolutely special people caring for them every day in our hospitals.

Authors

Peter Kidd studied pharmacy at Otago University, New Zealand and Queens University, Belfast. Whilst working as a hospital pharmacist in Wellington, London, Jeddah and Ireland he continued to study computer science and technology management. It was during this immersion in technology management that he was introduced to the Toyota Production System. He continued to explore improvement science through collaboration with Leading Edge, and latterly Simon Dodds. These relationships have been instrumental in progressing improvement projects within the hospital. His current roles are Lead for Pharmacy Clinical Services at University Hospital Galway, Adjunct Lecturer in Prescribing Science at National University of Ireland, Galway and Local Project Implementation Lead for the National Cancer Information System.

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Sponsor

Simon Dodds studied medicine and digital systems engineering before following a career in general and then vascular surgery. In 1999, he was appointed as a consultant surgeon at Good Hope Hospital in North Birmingham and applied his skills as an engineer and a clinician in the redesign of the vascular surgery clinic and the leg ulcer service. In 2004, the project was awarded a national innovation award for service improvement. This led to the design, development and dissemination of Health Care Systems Engineering (HCSE) programme.

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